

Final Project Step 2: Analysis Plan

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Research Question

Do AI-enabled medical devices that reach the market without reported clinical validation get recalled more often, or generate more adverse event reports, than devices backed by clinical testing?

Most AI devices are cleared through the FDA's 510(k) pathway, which does not require testing on patients. I expect devices without clinical validation to have more recalls and adverse event reports. If so, that would suggest the FDA should require more evidence before clearance.

Data

I start with the FDA's list of about 1,500 AI-enabled devices, which gives each device's decision date, submission number, company, and specialty. Each device's submission number and product code also appear in other FDA databases, so I can match it to:

- 510(k) clearance records: review pathway and review time
- Device classification file: risk class (I to III)
- Recall database: whether and when the device was recalled
- MAUDE adverse event database: number of problem reports
- 510(k) summary documents: whether the company ran a clinical study
- SEC company ticker list: whether the manufacturer is publicly traded

The result is one row per device. The outcomes are recall (yes/no) and adverse event count. The main variable is clinical validation, coded yes if the summary reports a study on patients or patient data. I will check this coding against about 100 summaries coded by hand. Controls are risk class, specialty, public company status, review time, and years on market.

Methodology

My intended methodology revolves more around testing. I will first explore the data to see how often AI devices are cleared with clinical validation and whether validated and non-validated devices already look different in their recall and adverse event records. Because those raw differences could be explained by other factors, such as risk class, specialty, or how long a device has been on the market, I will then use regression models that hold those factors constant. Each outcome gets a model suited to its type: recall is a yes/no outcome, while adverse events are counts. Finally, I intend to check whether the results hold up under a different modeling approach. The steps are:

1. Exploratory analysis

Describe how common clinical validation is across years, specialties, and risk classes, and compare recall rates and adverse event counts between validated and non-validated devices.

2. Recall model

A logistic regression will estimate the odds that a device is recalled, based on clinical validation and the controls.

3. Adverse event model

A negative binomial regression will model adverse event counts, since they are spread too widely for a Poisson model. Years on market is an offset, so the model compares reports per year:

$$\log(\mu_i) = \beta_0 + \beta_1 \text{Validated}_i + \gamma'X_i + \log(\text{Years}_i)$$

4. Robustness check

If time allows, a Cox model of time to first recall will account for newer devices having had less time to be recalled.

All analysis will be done in Python (pandas, pdfplumber, statsmodels, lifelines, matplotlib).

Analysis Steps

1. Download and clean the FDA and SEC data.
2. Match each device to its clearance, classification, recall, and adverse event records.
3. Extract text from the 510(k) summaries and code clinical validation.
4. Explore the data with summary tables and plots.
5. Fit the logistic and negative binomial models.
6. Run robustness checks and write up the results.